

Chapter 6

Distributions

Distributions are used to define stochastic nodes using the $\sim$ operator. The distributions defined in the `bugs` module are listed in table 6.1 (real-valued distributions), 6.2 (discrete-valued distributions), and 6.3 (multivariate distributions).

Some distributions have restrictions on the valid parameter values, and these are indicated in the tables. If a Distribution is given invalid parameter values when evaluating the log-likelihood, it returns $-\infty$. When a model is initialized, all stochastic nodes are checked to ensure that the initial parameter values are valid for their distribution.

6.1 Distribution aliases

A distribution may optionally have an alias, which can be used in the model definition in place of the canonical name. Aliases are used to avoid confusion with other statistical software in which distributions may have different names. Table 6.4 shows the distributions in the `bugs` module with an alias.

Name	Usage	Density	Lower	Upper
Beta	<code>dbeta(a,b)</code> $a > 0, b > 0$	$\frac{x^{a-1}(1-x)^{b-1}}{\beta(a,b)}$	0	1
Chi-square	<code>dchisqr(k)</code> $k > 0$	$\frac{x^{\frac{k}{2}-1} \exp(-x/2)}{2^{\frac{k}{2}} \Gamma(\frac{k}{2})}$	0	
Double exponential	<code>ddexp(mu,tau)</code> $\tau > 0$	$\tau \exp(-\tau x - \mu)/2$		
Exponential	<code>dexp(lambda)</code> $\lambda > 0$	$\lambda \exp(-\lambda x)$	0	
F	<code>df(n,m)</code> $n > 0, m > 0$	$\frac{\Gamma(\frac{n+m}{2})}{\Gamma(\frac{n}{2})\Gamma(\frac{m}{2})} \left(\frac{n}{m}\right)^{\frac{n}{2}} x^{\frac{n}{2}-1} \left\{1 + \frac{nx}{m}\right\}^{-\frac{(n+m)}{2}}$	0	
Gamma	<code>dgamma(r, lambda)</code> $\lambda > 0, r > 0$	$\frac{\lambda^r x^{r-1} \exp(-\lambda x)}{\Gamma(r)}$	0	
Generalized gamma	<code>dgen.gamma(r,lambda,b)</code> $\lambda > 0, b > 0, r > 0$	$\frac{b \lambda^{br} x^{br-1} \exp\{-(\lambda x)^b\}}{\Gamma(r)}$	0	
Logistic	<code>dlogis(mu, tau)</code> $\tau > 0$	$\frac{\tau \exp\{(x - \mu)\tau\}}{[1 + \exp\{(x - \mu)\tau\}]^2}$		
Log-normal	<code>dlnorm(mu,tau)</code> $\tau > 0$	$\left(\frac{\tau}{2\pi}\right)^{\frac{1}{2}} x^{-1} \exp\left\{-\tau(\log(x) - \mu)^2/2\right\}$	0	
Noncentral Chi-square	<code>dnchisqr(k, delta)</code> $k > 0, \delta \geq 0$	$\sum_{r=0}^{\infty} \frac{\exp(-\frac{\delta}{2})(\frac{\delta}{2})^r}{r!} \frac{x^{(k/2+r-1)} \exp(-\frac{x}{2})}{2^{(k/2+r)} \Gamma(\frac{k}{2}+r)}$	0	
Normal	<code>dnorm(mu,tau)</code> $\tau > 0$	$\left(\frac{\tau}{2\pi}\right)^{\frac{1}{2}} \exp\{-\tau(x - \mu)^2/2\}$		
Pareto	<code>dpar(alpha, c)</code> $\alpha > 0, c > 0$	$\alpha c^\alpha x^{-(\alpha+1)}$	c	
Student t	<code>dt(mu,tau,k)</code> $\tau > 0, k > 0$	$\frac{\Gamma(\frac{k+1}{2})}{\Gamma(\frac{k}{2})} \left(\frac{\tau}{k\pi}\right)^{\frac{1}{2}} \left\{1 + \frac{\tau(x-\mu)^2}{k}\right\}^{-\frac{(k+1)}{2}}$		
Uniform	<code>dunif(a,b)</code> $a < b$	$\frac{1}{b-a}$	a	b
Weibull	<code>dweib(v, lambda)</code> $v > 0, \lambda > 0$	$v \lambda x^{v-1} \exp(-\lambda x^v)$	0	

Table 6.1: Univariate real-valued distributions in the `bugs` module

Name	Usage	Density	Lower	Upper
Beta	<code>dbetabin(a, b, n)</code>	$\binom{a+x-1}{x} \binom{b+n-x-1}{n-x} \binom{a+b+n-1}{n}^{-1}$	0	n
binomial	$a > 0, b > 0, n \in \mathbb{N}^*$			
Bernoulli	<code>dbern(p)</code> $0 < p < 1$	$p^x (1-p)^{1-x}$	0	1
Binomial	<code>dbin(p, n)</code> $0 < p < 1, n \in \mathbb{N}^*$	$\binom{n}{x} p^x (1-p)^{n-x}$	0	n
Categorical	<code>dcat(pi)</code> $\pi \in (\mathbb{R}^+)^N$	$\frac{\pi_x}{\sum_i \pi_i}$	1	N
Noncentral hypergeometric	<code>dhyper(n1, n2, m1, psi)</code> $0 \leq n_i, 0 < m_1 \leq n_+$	$\frac{\binom{n_1}{x} \binom{n_2}{m_1-x} \psi^x}{\sum_i \binom{n_1}{i} \binom{n_2}{m_1-i} \psi^i}$	$\max(0, n_+ - m_1)$	$\min(n_1, m_1)$
Negative binomial	<code>dnegbin(p, r)</code> $0 < p \leq 1, r \geq 0$	$\binom{x+r-1}{x} p^r (1-p)^x$	0	
Poisson	<code>dpois(lambda)</code> $\lambda > 0$	$\frac{\exp(-\lambda) \lambda^x}{x!}$	0	

Table 6.2: Discrete univariate distributions in the **bugs** module

Name	Usage	Density
Dirichlet	<code>p ~ ddirch(alpha)</code> $\alpha_j \geq 0$	$\Gamma(\sum_i \alpha_i) \prod_j \frac{p_j^{\alpha_j-1}}{\Gamma(\alpha_j)}$
Multivariate normal	<code>x ~ dmnorm(mu, Omega)</code> Ω positive definite	$\left(\frac{ \Omega }{2\pi}\right)^{\frac{1}{2}} \exp\{-(x-\mu)^T \Omega (x-\mu)/2\}$
Wishart	<code>Omega ~ dwish(R, k)</code> R $p \times p$ pos. def., $k \geq p$	$\frac{ \Omega ^{(k-p-1)/2} R ^{k/2} \exp\{-\text{Tr}(R\Omega/2)\}}{2^{pk/2} \Gamma_p(k/2)}$
Multivariate Student t	<code>x ~ dmt(mu, Omega, k)</code> Ω pos. def.	$\frac{\Gamma\{(k+p)/2\}}{\Gamma(k/2)(n\pi)^{p/2}} \Omega ^{1/2} \left\{1 + \frac{1}{k}(x-\mu)^T \Omega (x-\mu)\right\}^{-\frac{(k+p)}{2}}$
Multinomial	<code>x ~ dmulti(pi, n)</code> $\sum_j x_j = n$	$n! \prod_j \frac{\pi_j^{x_j}}{x_j!}$

Table 6.3: Multivariate distributions in the **bugs** module

Distribution	Canonical name	Alias	Compatible with
Binomial	<code>dbin</code>	<code>dbinom</code>	R
Chi-square	<code>dchisqr</code>	<code>dchisq</code>	R
Negative binomial	<code>dnegbin</code>	<code>dnbinom</code>	R
Weibull	<code>dweib</code>	<code>dweibull</code>	R
Dirichlet	<code>ddirch</code>	<code>ddirich</code>	OpenBUGS

Table 6.4: Distributions with aliases in **bugs** module